Matryoshka Representation Learning (MRL) enables each of the first m dimensions of the embedding vector, $z_{1:m} \in \mathbb{R}^m$ to be independently capable of being a transferable and general purpose representation of the datapoint x. We obtain z using a deep neural network $F(\cdot\,;\theta_F)\colon \mathcal{X} \to \mathbb{R}^d$ parameterized by learnable weights θ_F, i.e., $z \coloneqq F(x;\theta_F)$. The multi-granularity is captured through the set of the chosen dimensions $\mathcal{M}$, that contains less than $\log(d)$ elements, i.e., $|\mathcal{M}| \leq \lfloor \log(d) \rfloor$. The usual set $\mathcal{M}$ consists of consistent halving until the representation size hits a low information bottleneck. We discuss the design choices in Section 4 for each of the representation learning settings.

For the ease of exposition, we present the formulation for fully supervised representation learning via multi-class classification. Matryoshka Representation Learning modifies the typical setting to become a multi-scale representation learning problem on the same task. For example, we train ResNet50 [29] on ImageNet-1K [76] which embeds a 224×224 pixel image into a $d = 2048$ representation vector and then passed through a linear classifier to make a prediction, $\hat{y}$ among the $L = 1000$ labels. For MRL, we choose $\mathcal{M} = \{8, 16, \ldots, 1024, 2048\}$ as the nesting dimensions.

Suppose we are given a labelled dataset $\mathcal{D} = \{(x_1, y_1), \ldots, (x_N, y_N)\}$ where $x_i \in \mathcal{X}$ is an input point and $y_i \in [L]$ is the label of x_i for all $i \in [N]$. MRL optimizes the multi-class classification loss for each of the nested dimension $m \in \mathcal{M}$ using standard empirical risk minimization using a separate linear classifier, parameterized by $\mathbf{W}^{(m)} \in \mathbb{R}^{L \times m}$. All the losses are aggregated after scaling with their relative importance $(c_m \geq 0)_{m \in \mathcal{M}}$ respectively. That is, we solve

$$\min_{\left\{\mathbf{W}^{(m)}\right\}_{m\in\mathcal{M}},\ \theta_F} \frac{1}{N} \sum_{i \in [N]} \sum_{m \in \mathcal{M}} c_m \cdot \mathcal{L}\left(\mathbf{W}^{(m)} \cdot F(x_i;\theta_F)_{1:m}\ ;\ y_i\right) , \tag{1}$$

where $\mathcal{L}\colon \mathbb{R}^L \times [L] \to \mathbb{R}_+$ is the multi-class softmax cross-entropy loss function. This is a standard optimization problem that can be solved using sub-gradient descent methods. We set all the importance scales, $c_m = 1$ for all $m \in \mathcal{M}$; see Section 5 for ablations. Lastly, despite only optimizing for $O(\log(d))$ nested dimensions, MRL results in accurate representations, that interpolate, for dimensions that fall between the chosen granularity of the representations (Section 4.2).

We call this formulation as Matryoshka Representation Learning (MRL). A natural way to make this efficient is through weight-tying across all the linear classifiers, i.e., by defining $\mathbf{W}^{(m)} = \mathbf{W}_{1:m}$ for a set of common weights $\mathbf{W} \in \mathbb{R}^{L \times d}$. This would reduce the memory cost due to the linear classifiers by almost half, which would be crucial in cases of extremely large output spaces [89, 99]. This variant is called *Efficient* Matryoshka Representation Learning (MRL–E). Refer to Alg 1 and Alg 2 in Appendix A for the building blocks of Matryoshka Representation Learning (MRL).

Adaptation to Learning Frameworks. MRL can be adapted seamlessly to most representation learning frameworks at web-scale with minimal modifications (Section 4.1). For example, MRL's adaptation to masked language modelling reduces to MRL–E due to the weight-tying between the input embedding matrix and the linear classifier. For contrastive learning, both in context of vision & vision + language, MRL is applied to both the embeddings that are being contrasted with each other. The presence of normalization on the representation needs to be handled independently for each of the nesting dimension for best results (see Appendix C for more details).

4 Applications

In this section, we discuss Matryoshka Representation Learning (MRL) for a diverse set of applications along with an extensive evaluation of the learned multifidelity representations. Further, we showcase the downstream applications of the learned Matryoshka Representations for flexible large-scale deployment through (a) Adaptive Classification (AC) and (b) Adaptive Retrieval (AR).

4.1 Representation Learning

We adapt Matryoshka Representation Learning (MRL) to various representation learning setups (a) Supervised learning for vision: ResNet50 [29] on ImageNet-1K [76] and ViT-B/16 [22] on JFT-300M [85], (b) Contrastive learning for vision + language: ALIGN model with ViT-B/16 vision encoder and BERT language encoder on ALIGN data [46] and (c) Masked language modelling: BERT [19] on English Wikipedia and BooksCorpus [102]. Please refer to Appendices B and C for details regarding the model architectures, datasets and training specifics.

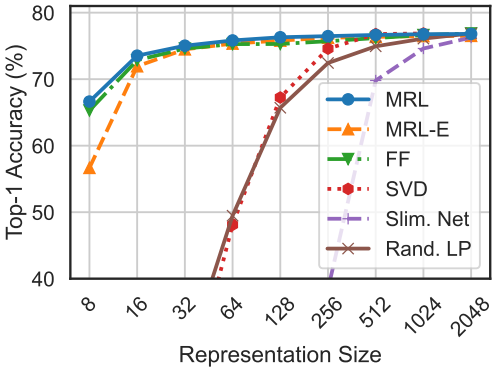

Figure 2: ImageNet-1K linear classification accuracy of ResNet50 models. MRL is as accurate as the independently trained FF models for every representation size.

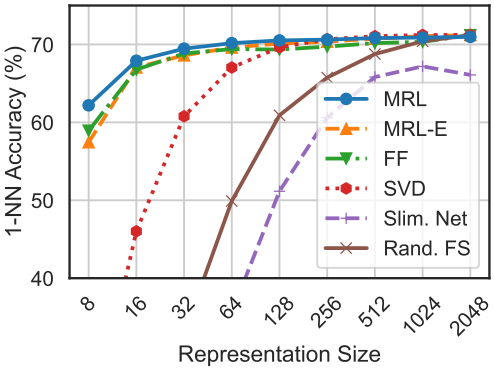

Figure 3: ImageNet-1K 1-NN accuracy of ResNet50 models measuring the representation quality for downstream task. MRL outperforms all the baselines across all representation sizes.

We do not search for best hyper-parameters for all MRL experiments but use the same hyper-parameters as the independently trained baselines. ResNet50 outputs a 2048-dimensional representation while ViT-B/16 and BERT-Base output 768-dimensional embeddings for each data point. We use $\mathcal{M} = \{8, 16, 32, 64, 128, 256, 512, 1024, 2048\}$ and $\mathcal{M} = \{12, 24, 48, 96, 192, 384, 768\}$ as the explicitly optimized nested dimensions respectively. Lastly, we extensively compare the MRL and MRL–E models to independently trained low-dimensional (fixed feature) representations (FF), dimensionality reduction (SVD), sub-net method (slimmable networks [100]) and randomly selected features of the highest capacity FF model.

In section 4.2, we evaluate the quality and capacity of the learned representations through linear classification/probe (LP) and 1-nearest neighbour (1-NN) accuracy. Experiments show that MRL models remove the dependence on $|\mathcal{M}|$ resource-intensive independently trained models for the coarse-to-fine representations while being as accurate. Lastly, we show that despite optimizing only for $|\mathcal{M}|$ dimensions, MRL models diffuse the information, in an interpolative fashion, across all the d dimensions providing the finest granularity required for adaptive deployment.

4.2 Classification

Figure 2 compares the linear classification accuracy of ResNet50 models trained and evaluated on ImageNet-1K. ResNet50–MRL model is at least as accurate as each FF model at every representation size in $\mathcal{M}$ while MRL–E is within 1% starting from 16-dim. Similarly, Figure 3 showcases the comparison of learned representation quality through 1-NN accuracy on ImageNet-1K (trainset with 1.3M samples as the database and validation set with 50K samples as the queries). Matryoshka Representations are up to 2% more accurate than their fixed-feature counterparts for the lower-dimensions while being as accurate elsewhere. 1-NN accuracy is an excellent proxy, at no additional training cost, to gauge the utility of learned representations in the downstream tasks.

We also evaluate the quality of the representations from training ViT-B/16 on JFT-300M alongside the ViT-B/16 vision encoder of the ALIGN model – two web-scale setups. Due to the expensive nature of these experiments, we only train the highest capacity fixed feature model and choose random features for evaluation in lower-dimensions. Web-scale is a compelling setting for MRL due to its relatively inexpensive training overhead while providing multifidelity representations for downstream tasks. Figure 4, evaluated with 1-NN on ImageNet-1K, shows that all the MRL models for JFT and ALIGN are highly accurate while providing an excellent cost-vs-accuracy trade-off at lower-dimensions. These experiments show that MRL seamlessly scales to large-scale models and web-scale datasets while providing the otherwise prohibitively expensive multi-granularity in the process. We also have similar observations when pretraining BERT; please see Appendix D.2 for more details. Our experiments also show that post-hoc compression (SVD), linear probe on random features, and sub-net style slimmable networks drastically lose accuracy compared to MRL as the representation size decreases. Finally, Figure 5 shows that, while MRL explicitly optimizes $O(\log(d))$ nested representations – removing the $O(d)$ dependence [73] –, the coarse-to-fine grained information is interpolated across all d dimensions providing highest flexibility for adaptive deployment.

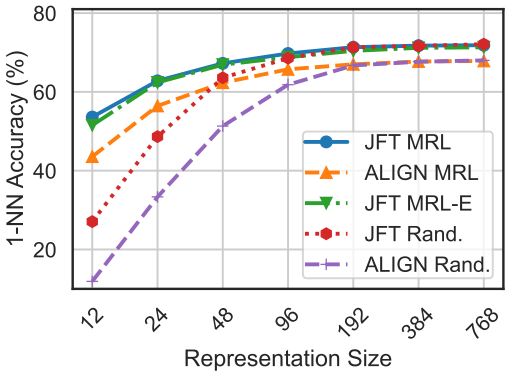

Figure 4: ImageNet-1K 1-NN accuracy for ViT-B/16 models trained on JFT-300M & as part of ALIGN. MRL scales seamlessly to web-scale with minimal training overhead.

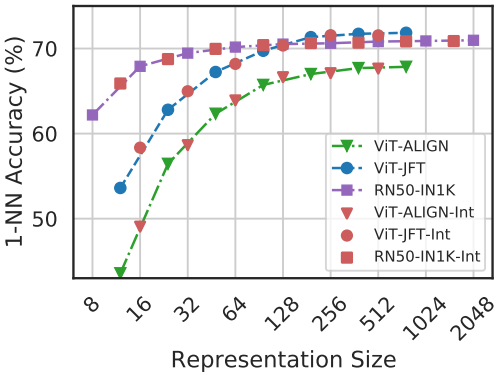

Figure 5: Despite optimizing MRL only for $O(\log(d))$ dimensions for ResNet50 and ViT-B/16 models; the accuracy in the intermediate dimensions shows interpolating behaviour.

4.2.1 Adaptive Classification

The flexibility and coarse-to-fine granularity within Matryoshka Representations allows model cascades [90] for Adaptive Classification (AC) [28]. Unlike standard model cascades [95], MRL does not require multiple expensive neural network forward passes. To perform AC with an MRL trained model, we learn thresholds on the maximum softmax probability [33] for each nested classifier on a holdout validation set. We then use these thresholds to decide when to transition to the higher dimensional representation (e.g $8 \rightarrow 16 \rightarrow 32$) of the MRL model. Appendix D.1 discusses the implementation and learning of thresholds for cascades used for adaptive classification in detail.

Figure 6 shows the comparison between cascaded MRL representations (MRL–AC) and independently trained fixed feature (FF) models on ImageNet-1K with ResNet50. We computed the expected representation size for MRL–AC based on the final dimensionality used in the cascade. We observed that MRL–AC was as accurate, 76.30%, as a 512-dimensional FF model but required an expected dimensionality of ~ 37 while being only 0.8% lower than the 2048-dimensional FF baseline. Note that all MRL–AC models are significantly more accurate than the FF baselines at comparable representation sizes. MRL–AC uses up to $\sim 14\times$ smaller representation size for the same accuracy which affords computational efficiency as the label space grows [89]. Lastly, our results with MRL–AC indicate that instances and classes vary in difficulty which we analyze in Section 5 and Appendix J.

4.3 Retrieval

Nearest neighbour search with learned representations powers a plethora of retrieval and search applications [15, 91, 11, 66]. In this section, we discuss the image retrieval performance of the pretrained ResNet50 models (Section 4.1) on two large-scale datasets ImageNet-1K [76] and ImageNet-4K. ImageNet-1K has a database size of ~1.3M and a query set of 50K samples uniformly spanning 1000 classes. We also introduce ImageNet-4K which has a database size of ~4.2M and query set of ~200K samples uniformly spanning 4202 classes (see Appendix B for details). A single forward pass on ResNet50 costs 4 GFLOPs while exact retrieval costs 2.6 GFLOPs per query for ImageNet-1K. Although retrieval overhead is 40% of the total cost, retrieval cost grows linearly with the size of the database. ImageNet-4K presents a retrieval benchmark where the exact search cost becomes the computational bottleneck (8.6 GFLOPs per query). In both these settings, the memory and disk usage are also often bottlenecked by the large databases. However, in most real-world applications exact search, $O(dN)$, is replaced with an approximate nearest neighbor search (ANNS) method like HNSW [62], $O(d\log(N))$, with minimal accuracy drop at the cost of additional memory overhead.

The goal of image retrieval is to find images that belong to the same class as the query using representations obtained from a pretrained model. In this section, we compare retrieval performance using mean Average Precision @ 10 (mAP@10) which comprehensively captures the setup of relevant image retrieval at scale. We measure the cost per query using exact search in MFLOPs. All embeddings are unit normalized and retrieved using the L2 distance metric. Lastly, we report